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Harbin Institute of Technology

Published: 24 March 2025

[Citation in BibTeX format](#)

IUI '25: 30th International Conference on
Intelligent User Interfaces Companion
March 24 - 27, 2025
Cagliari, Italy

Conference Sponsors:
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SLIMO CITY: A Bio-HCI Game System Amplifying Slime Mold Behaviors through Urban Visualization

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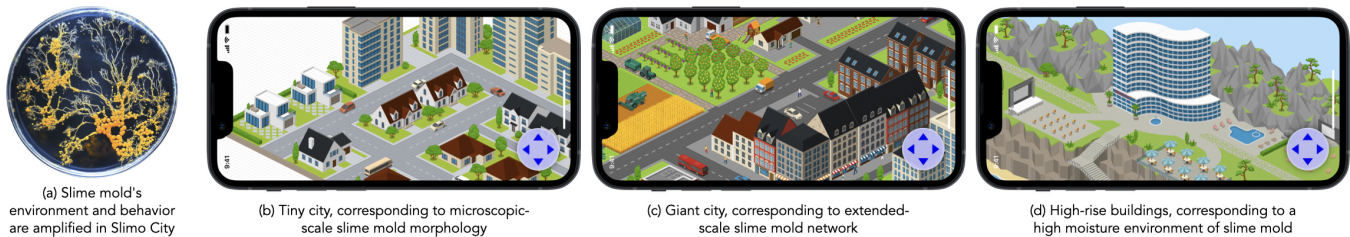


Figure 1: The display of slime mold and three game interfaces of Slimo City design via Icongrams.

Abstract

We present SLIMO CITY, a bio-driven urban visualization game that translates slime mold (*Physarum polycephalum*) behavior into dynamic city development. By mapping biological responses to urban elements, our system transforms microscopic activities into an intuitive urban-scale visualization. The system monitors slime mold's growth patterns and environmental responses through integrated sensing mechanisms, translating these into three key urban parameters: city scale, color palette, and building height. This work contributes to Bio-HCI research by demonstrating how biological behaviors can drive interactive visualization systems while making complex natural processes accessible through familiar urban metaphors.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Visualization design and evaluation methods**; **Interface design prototyping**.

Keywords

Bio-HCI, Slime Mold, Living Interfaces, Real-time Visualization, Game

ACM Reference Format:

Mengyao Guo and Jinda Han. 2025. SLIMO CITY: A Bio-HCI Game System Amplifying Slime Mold Behaviors through Urban Visualization. In *30th*

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IUI Companion '25, Cagliari, Italy

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ACM ISBN 979-8-4007-1409-2/25/03

<https://doi.org/10.1145/3708557.3716346>

International Conference on Intelligent User Interfaces Companion (IUI Companion '25), March 24–27, 2025, Cagliari, Italy. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3708557.3716346>

1 Introduction

The intersection of Bio-HCI (Human-Computer Interaction) [2, 4, 14] and game design [7, 16] opens a unique door for exploring biological systems through digital interfaces [13]. While Bio-HCI integrates living organisms as active interaction elements [18, 19], games serve as an intuitive platform for visualizing complex biological behaviors [5]. This convergence enables us to move beyond traditional human-machine interfaces toward systems that interpret and amplify natural phenomena engagingly [11, 12]. Drawing from animistic perspectives that recognize the inherent intelligence and communication methods of all living beings, we position our work within a framework that acknowledges the equal significance of all life forms. Each organism, from the simplest to the most complex, possesses unique ways of interacting with its environment - a principle particularly evident in slime molds (*Physarum polycephalum*). By amplifying and interpreting these natural communication methods, we can better understand and learn from these biological systems.

Our research focuses on slime molds, an organism that emerges as an ideal candidate at this intersection. Despite its simple single-celled structure, slime mold exhibits intelligent environmental responses and network optimization behaviors, offering rich potential for gamified interpretation [15]. These sophisticated biological behaviors [20], typically occurring at microscopic scales too subtle for direct human observation, become accessible through SLIMO CITY's urban-scale visualization system (see Figure 1).

The gamification of biological behavior through urban metaphors transforms microscopic biological processes into observable urban patterns [1, 8]. This approach renders invisible processes visible and enables real-time interaction with living systems while maintaining

user engagement through familiar design elements. SLIMO CITY establishes a bio-centered methodology [17] by positioning biological elements as central actors in the interaction system, where natural behaviors actively drive the design (see Figure 2). Through synthesizing Bio-HCI, game design, and biological observation, our project advances the understanding of natural intelligence while opening new directions for bio-inspired technology development.



Figure 2: The game interfaces of Slimo City show the city’s growth in response to slime mold growth, designed via Icoograms and Figma.

2 Related Works

Our work builds upon three key areas of previous research: bio-sensing interfaces, slow computing paradigms, and biological computing systems. SLIMO CITY uniquely combines these aspects by using game mechanics to amplify and visualize biological behavior while maintaining the natural temporal aspects of slime mold responses.

The study of alternative temporalities in the system of biological-computational highlights fundamental tensions between digital immediacy and natural processes. Ikeya’s Algal Relay Computer (ARC) directly addressed these tensions by incorporating microalgae growth into computational processes (see Figure 3a), demonstrating how biological time scales can be meaningfully integrated into technological systems [6, 21]. This consideration of biological temporality has shaped new approaches to human-microbe interaction design. Chen et al.’s Nukabot (see Figure 3b) explored the development of human-microbe relationships through a technologically enhanced Japanese pickling system [3, 10]. Their work demonstrated how voice interaction and anthropomorphic design elements foster affective relationships with microbes through daily



Figure 3: The related works.

care practices. Complementing these developments in biological interaction, advances in bio-sensing and visualization have opened new possibilities for understanding biological systems. Kim et al.’s cMaSK exemplified how biological signals can inform technological design through conformable sensor interfaces that adapt to natural forms [9]. Their research illustrates how technological systems can embody biological principles of adaptability and sensitivity while making invisible processes observable (see Figure 3c).

3 SLIMO CITY: As an Amplifier

SLIMO CITY builds upon these foundations by synthesizing real-time biological sensing and visualization into a unified framework where biological behaviors drive system evolution through accessible urban metaphors (see Figure 4).

3.1 Technical Implementation

We developed a sensing system for this game that combines computer vision, light detection, and environmental monitoring to capture slime mold’s dynamic behaviors.

OpenCV: Our growth detection system utilizes [OpenCV](#) with a USB camera positioned above the slime mold culture (see Figure 5). Real-time video capture and continuous tracking of the organism’s morphological changes through blob detection and contour analysis when the slime mold consumes the food humans feed. This system monitors growth rate, expansion patterns, and size changes by analyzing area coverage through pixel counting and tracking contour evolution in three steps:

- Image pre-processing (Grayscale conversion and threshold adjustment): The first step is to prepare the images of the slime mold so the system can analyze them more easily. To

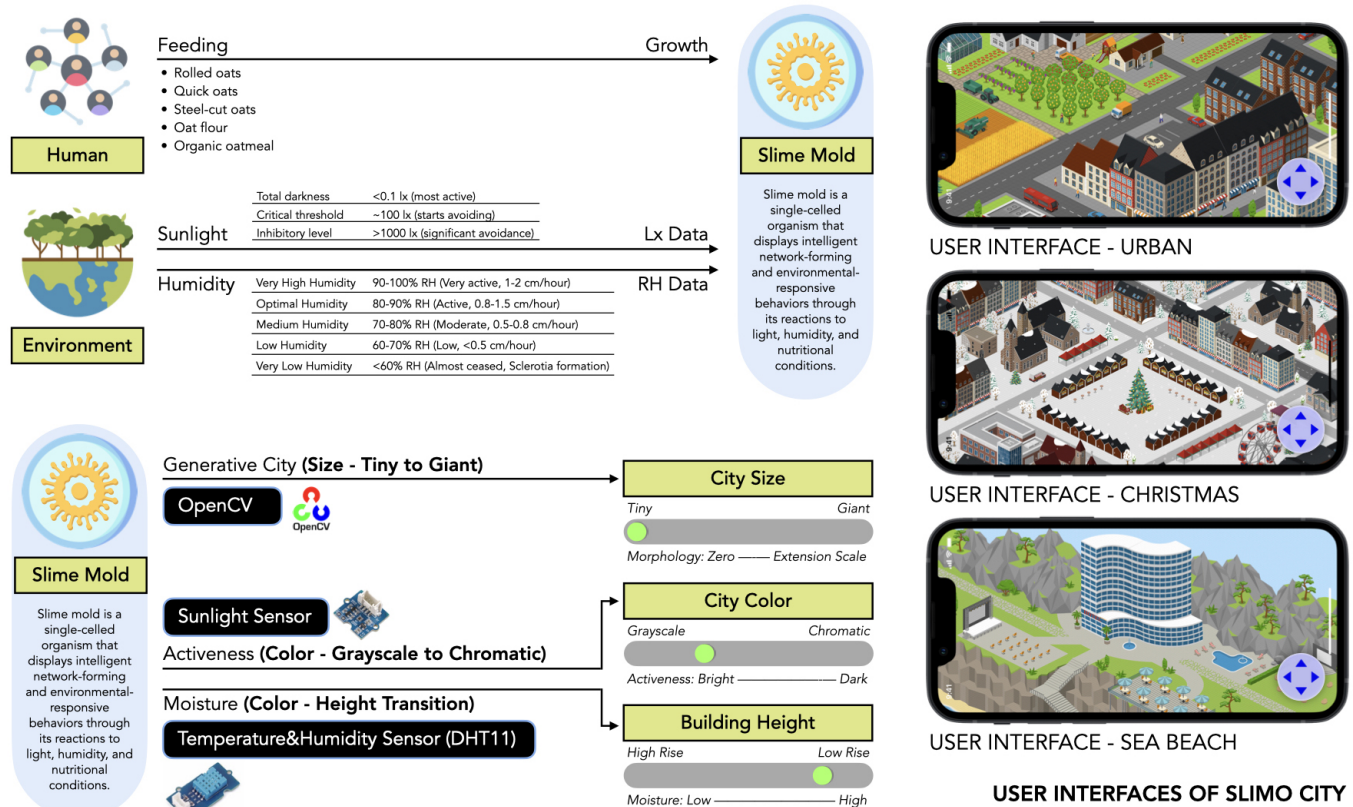


Figure 4: The workflow of Slimo City.

do this, the system converts the video frames into black-and-white (grayscale) images. This removes unnecessary color details, making it easier to focus on the important parts of the image. After that, the system adjusts the brightness levels to create a clear contrast between the slime mold and the background. This step helps highlight the slime mold and prepares the image for further analysis.

- **Contour detection (Extraction of slime mold boundaries):** Next, the system looks for the edges of the slime mold (contours). These edges show the shape and size of the slime mold as it grows. By detecting these boundaries, the system can understand how the slime mold is spreading and changing over time. This process is like tracing the outline of the slime mold in each image.
- **Area calculation (Computation of the maximum contour area):** Once the edges are found, the system calculates how much space the slime mold is taking up. It does this by measuring the total area inside the outline. By tracking changes in this area, the system can see how fast the slime mold is growing and how active it is. This information is then used to create the game's city visuals, where the size and density of the city change based on how the slime mold grows.

These growth parameters directly drive the city's scale and urban density in our visualization system. The processed growth data

is stored in a *growth_data.json* file, containing timestamped size variations for visualization mapping.

Arduino Mega with Sensors: For data collection, we utilize an **Arduino Mega** microcontroller as the central hub, synchronizing inputs from the **sunlight sensor** and **temperature & humidity sensor (DHT11)** through direct pin connections (see Figure 5).

A sunlight sensor monitors ambient light conditions across critical thresholds for activity detection. The system measures light levels ranging from total darkness (<0.1 lx, indicating peak activity) to inhibitory levels (>1000 lx, triggering significant avoidance). These measurements inform the visualization's color scheme, transitioning from grayscale to chromatic based on activity levels. Environmental response monitoring utilizes DHT11 to track humidity conditions. The system distinguishes five humidity ranges: from very high (90-100% RH) indicating peak activity, to very low (<60% RH) signifying minimal activity. These humidity measurements drive building height transitions in urban visualization, establishing a direct relationship between environmental conditions and architectural form.

3.2 Game Visualization

In terms of our game visualization, the Unity-based visualization system translates biological data into a dynamic urban environment through three interconnected parameters. The system initializes

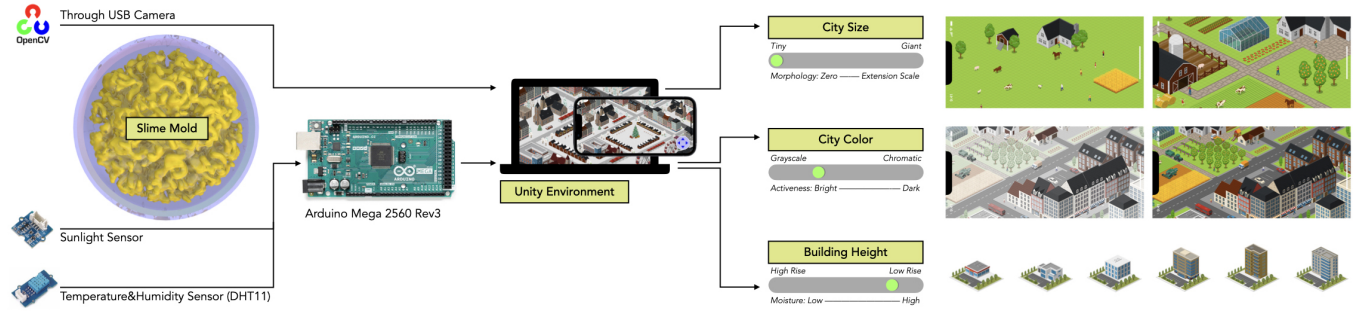


Figure 5: The data communication of Slimo City.

with a rural setting, featuring farmland with essential agricultural elements such as barns, houses, and greenhouses.

Scale transformation directly responds to slime mold's growth patterns, captured through OpenCV measurements. The visualization progresses from an initial agricultural landscape to increasingly complex urban developments. This evolution transitions from scattered farm buildings to comprehensive urban environments, introducing residential areas, commercial districts, and diverse architectural styles. The system maintains visual continuity by preserving original elements while integrating new urban features, including roads, public spaces, and varied building types.

Environmental activity levels, detected by the sunlight sensor, govern the visualization's color palette. The scene adopts a muted grayscale across all elements during low-activity periods. As slime mold's activity increases, the environment transitions to vibrant chromatic representations, featuring bright greens in vegetation, warm tones in buildings, and diverse colors in urban features. This color evolution provides immediate feedback about the organism's current state.

Building height variations, determined by humidity readings from the DHT11 sensor, create dynamic vertical development. The system implements a graduated scale from single-story rural structures to high-rise urban developments. Each height stage maintains architectural consistency while introducing more complex structural elements, including detailed windows, balconies, and roof variations that reflect the progression from rural to urban density.

4 Conclusion and Future Work

SLIMO CITY demonstrates a novel approach to Bio-HCI by translating microscopic biological behaviors into an interactive urban-scale visualization system. By integrating computer vision, environmental sensing, and real-time visualization, our system creates an accessible interface for observing and interacting with slime mold's intelligent behaviors.

While SLIMO CITY can translate slime mold growth into urban-scale visualization, two fundamental limitations remain. First, the system cannot yet replicate the organism's specific growth patterns in the city's layout, as urban development follows predetermined patterns rather than emerging from biological forms. Second, our use of standardized ICOGRAMS assets (see Figure 6), while ensuring visual consistency, constrains the creation of unique architectural expressions that could better reflect the organism's behavior.

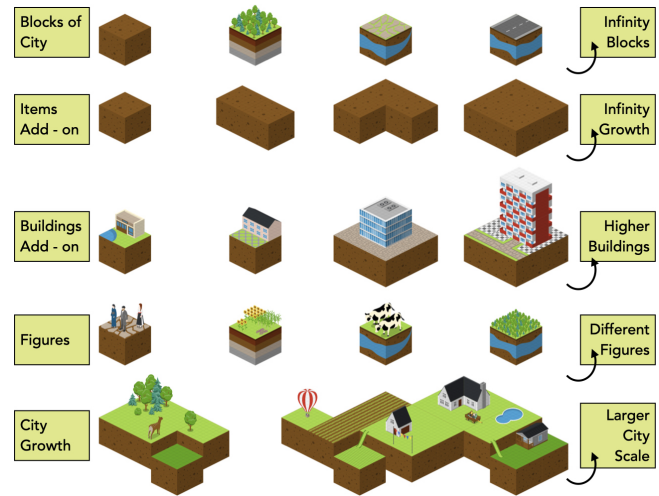


Figure 6: The blocks of Slimo City, designed via IcoGrams.

Our future development will focus on two major improvements. First, we plan to integrate generative AI techniques to enable morphological development that directly mirrors slime mold's growth patterns. This AI system will analyze the organism's structure in real-time, generating corresponding urban layouts that maintain architectural coherence and biological authenticity. Second, we will implement wireless communication through an HC-05 RF Bluetooth Transceiver Module (RS232), eliminating physical constraints between sensors and the visualization system.

Acknowledgments

The authors would like to thank the organizers of the workshop "Towards New Possibilities of Bio-HCI: Exploring Future Scenarios of Bio-HCI Innovation" at Chinese CHI 2024 by Christiane Herr, Haorui Tian, and Umaru Mohammed from the Southern University of Science and Technology, Shenzhen, China. Their valuable feedback and support contributed significantly to this work.

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